Sound and Deterministic Quadric Arrangements on a Rational Ellipsoid with Machine-Checkable Certificates

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Abstract

We present a deterministic arrangement kernel for rational quadric surfaces on a rational ellipsoid, built on exact rational and algebraic arithmetic. Intersection curves are pulled back to the plane through a rational stereographic atlas of the ellipsoid, and two-dimensional algebraic curve arrangements are computed with resultants, subresultants, and univariate root isolation. Every rejected input carries a classified `NoneReason` (a machine-readable label, not a per-input certificate); on our scoped benchmark all 378 single-chart failures are classified as unbounded charts (364) or confluent components (14), with zero unclassified cases. A deterministic atlas of six axial and twenty-four general poles reduces the residual all-chart rejection rate on a 2–3 plane family from 14.98% at coefficient height `H=1` to 0.64% at `H=3`, 0.020% at `H=10`, and 0.000% at `H=20`, an empirical `O(1/H)` trend rather than a theorem. For 20 mixed quartic pullbacks, the standard chart closes 14; the 6+24 chart search with a 3D face-point check and the single-safe-chart 2D lift both close and certify all 20, with strict loop-boundary 20/20 and last-resort 3D check 0/20. On our fixed-seed families, a 20-case three-quadric pilot is 20/20 via the hybrid path (single-chart plus exact rational linear-transform and basis fallbacks), certified and strict 20/20; a 100-case three-quadric stress test closes 100/100 with 100/100 certified and strict and no timeout. We also report 3/4/5/6-plane lifts in 20/20 cases each, certified and strict; seven or more planes are rejected by the certified API. Accepted outputs are independently checkable by a certificate checker. Here "certified" means that the output passes that checker (Euler, exact face signs, rotation system, and the strict loop-boundary invariants); it does not assert event completeness. A Lean-checked combinatorial core, extracted-data checks, and Mathlib proofs of the cusp sign and boundedness principle are included. We do not claim unconditional completeness or full formal verification of the arrangement kernel.

1. Introduction

1.1 Motivation

Implicit and CSG-based design pipelines rely on quadric primitives such as spheres, ellipsoids, cylinders, cones, and general quadrics. Boolean evaluation requires the arrangement of their boundary surfaces: a decomposition into vertices, edges, faces, and cells with consistent labels. Floating-point implementations are fragile at tangencies, coplanar contacts, shared components, and surfaces passing through a parameterization pole. Exact arithmetic removes rounding, but introduces expensive algebraic numbers and makes cross-environment determinism difficult.

This paper asks a narrower question than "solve arbitrary CSG": can we build a small, auditable arrangement kernel for rational quadrics that is exact, deterministic, fail-closed, and reproducible, with an explicit account of what it does not solve?

1.2 Problem and challenges

We consider a rational ellipsoid `E` and a finite set of rational quadrics. The kernel must compute their arrangement on `E`, lift the 2D arrangement to 3D boundary edges and faces, attach exact face labels, and reject unsupported or undecided configurations soundly. The certified API returns only outputs that pass the certificate gate.

The main difficulties are unbounded pullbacks (a chart pole on a curve), degeneracies (common components, ordinary cusps, vertical tangencies), algebraic-number ordering and membership, and determinism across machines and runs.

1.3 Approach

We use a rational multi-chart atlas:

1. Parameterize `E` by a rational stereographic map from a pole `P`.
2. Pull each quadric back to a polynomial `F_P(u,v)` of degree at most four.
3. If the excluded pole is not on the quadric, the pullback is bounded; otherwise try another chart from a deterministic atlas of six axial and twenty-four general poles.
4. Compute the 2D arrangement with resultants, subresultants, and univariate root isolation.
5. Lift the accepted single-chart arrangement to 3D and attach exact face labels. The current implementation does **not** stitch different charts; the atlas is a candidate pool, and each successful candidate is reconstructed in one chart.
6. If no chart works, or a degeneracy cannot be resolved, return `Rejected(None(reason))` with a classified `NoneReason`.

Two reconstruction modes share the same 6+24 candidate pool. The **single-chart 2D lift** uses the certified 2D labels and compactifies the unbounded face at the pole. The **3D-face-check mode** lifts the face sample points to `E` and verifies their signs in 3D; it is a validation mode, not a multi-chart stitch. Neither mode combines different charts in one output. Genuine multi-chart stitching is future work (OP2).

1.4 Contributions

ID	Contribution	Evidence
C1	All geometric predicates and topology use exact rational/algebraic arithmetic; f64 is confined to candidate generation.	code, determinism tests
C2	Every rejection has a classified `NoneReason` (a label, not a per-input certificate); 378/378 classified, zero unclassified on the scoped benchmark.	artifact, Lean totality model
C3	Deterministic 6+24 pole atlas over classical stereographic charts on the 2–3 plane family; residual all-chart rejection 14.98% -> 0.64% -> 0.020% -> 0.000% for `H=1,3,10,20` (empirical).	artifact, ablation
C4	Quartic pullbacks: 2-quadric 20-case benchmark: standard 14/20; 6+24 chart search with 3D face-point check 20/20 certified; single-chart 2D lift 20/20 certified + strict 20/20; last-resort 3D check 0/20. 3-quadric 20-case pilot: 20/20 hybrid, certified + strict 20/20. A 100-case 3-quadric stress test closes 100/100, certified + strict 100/100, with no timeout under the 300 s/case harness; 61 cases use the primary single-chart path and 39 use exact rational linear-transform/basis fallbacks. The claim remains empirical on the fixed-seed family, not an unconditional three-quadric theorem.	artifact
C5	A specific exact membership test (dynamic zero test) without bivariate GCD or algebraic-number arithmetic; counterexample regressions. The general avoidance of irrational numbers is prior work [Caravantes–Gonzalez-Vega 2008].	tests
C6	`GL(2)`-invariant Segre key and exact comparison for the handled pairs (not a new QSIC classification).	tests, related work
C7	Deterministic cusp merging via the exact branch-side sign `delta^2 = -x^3 T_3/(3 F_vv^4)`.	Rust regression, Mathlib proof
C8	Cost-aware algebraic-number arithmetic with guards.	timing tests
C9	Certified 3D lift for an ellipsoid plus up to six planes: 3/4/5/6-plane 20/20 cases pass a scoped certificate and strict check; max `V/E/F` is 12/18/8 (3), 20/32/14 (4, C1), 30/50/22 (5), 42/72/32 (6). Seven or more planes are rejected by the certified API.	artifact, gate
C10	Accepted outputs are independently checkable by a certificate checker (Euler, exact face signs, rotation system).	checker, tamper tests
C11	Lean-checked combinatorial core, Rust-to-Lean extraction, and Mathlib proofs of the cusp sign and boundedness principle.	Lean development (not included)
C12	Deterministic, reproducible artifacts (SHA-256, CI, fixed seeds).	artifacts, CI

1.5 Positioning and non-claims

We do not claim a new arrangement algorithm or a new classification of quadric intersection curves. Prior work establishes exact 3D quadric arrangements, complete planar maps of quadric intersection curves, exact cell computation including degenerate cases, exact low-degree curve arrangements, general exact arrangement frameworks, exact boundary evaluation, isotopy-guaranteed implicit meshing, and a complete QSIC classification. Prior work also establishes exact quartic and higher-degree curve arrangements (Caravantes–Gonzalez-Vega 2007, 2008; Eigenwillig et al. 2006), arrangements on parametric surfaces and stereographic parameterizations (Berberich et al. 2007, 2010; Wang–Joe–Goldman 1997), exact spherical arrangements (Cazals–Loriot 2009; de Castro et al. 2009; Fogel 2009), formal-methods CSG (Tzoumas et al. 2015), and certifying algorithms (McConnell et al. 2011; Li–Passmore–Paulson 2019). Our contribution is the combination of guarantees: deterministic execution, fail-closed rejection certificates, a rational atlas for unbounded pullbacks, empirical completeness on a scoped benchmark, an independent certificate checker, and a partially formalized combinatorial core.

We do not claim unconditional completeness, a solution for four general quadrics, a certified solution for seven or more planes, full formal verification of the arrangement kernel, or that `O(1/H)` is a theorem.

1.6 Roadmap

Section 2 fixes scope and semantics. Section 3 describes the method. Section 4 states soundness and determinism. Section 5 reports the completeness benchmark. Section 6 covers 3D arrangements and applications. Section 7 states limitations. Section 8 positions the work. Section 9 documents reproducibility.

1.7 Scope and open problems

This paper focuses on a scoped class: rational quadric surfaces on a rational ellipsoid, with certified output on the stated benchmark. Complete resolution for arbitrary rational quadrics (no chart or cost assumption, all degeneracies) is not claimed. Open problems include a general proof of the strict loop-boundary certificate (the benchmark check passes), multi-chart stitching when no common safe chart exists, general four-quadric scenes, and unconditional termination and cost bounds. Lean coverage is partial and is treated as a verification aid, not as full formalization.

2. Scope and Semantics

2.1 Input and output

Input. A rational ellipsoid `E` and rational quadrics `Q_1,...,Q_n`. The base ellipsoid admits a rational stereographic parameterization. Coefficients are exact integers or rationals.

Output. A boundary complex on `E`: vertices with exact algebraic coordinates and incidence metadata; edges labeled by their supporting quadric; faces with a sign vector in `{-1,0,+1}^n`; a rotation system; component count and Euler data. The output is an arrangement kernel, not a full CSG solid.

2.2 Exact arithmetic and the f64 boundary

All predicates and combinatorial decisions use exact rational or algebraic arithmetic. Floating point is used only to generate candidate charts, seed isolating intervals, and estimate costs. A candidate is never accepted on a floating-point test. If an exact decision cannot be made within the cost budget, the result is `Rejected(None(reason))`.

2.3 Failure semantics

Every rejection carries a `NoneReason`: `NonBoundedChart`, `CommonComponent`, `TangentialIntersection`, `SingularPoint`, `FaceCycleInvalid`, `OutOfScope`, or `BudgetExceeded`. `OutOfScope` is used for inputs outside the validated API scope (for example seven or more planes) and is never a geometric failure. `BudgetExceeded` is the fail-closed result of the deterministic `SurveyBudget`; it is distinct from `CommonComponent`. `Undetermined` is a sentinel for a missing classification; the kernel must not return it for a known rejection. The Lean totality model mirrors the Rust constructors (`nonBounded`, `commonComponent`, `tangential`, `singular`, `faceCycleInvalid`, `outOfScope`, `budgetExceeded`, `undetermined`), and its classification proof covers the same finite set. On the scoped benchmark the single-chart diagnostic records 378 rejections (364 non-bounded, 14 confluent components) and zero unclassified cases. The product-geometry run over circles and ellipsoid-plane pullbacks has no general-position rejection. The Lean model proves that a classified `RejectReason` maps to a non-`undetermined` `NoneReason`.

2.4 Claim boundary

The paper claims soundness of accepted outputs on the scoped benchmark, practical completeness via the atlas, exact 3D arrangements for an ellipsoid plus up to six planes, and deterministic reproducible artifacts. It does not claim completeness for seven or more planes, four general quadrics, mesh fidelity, unconditional formal verification, or a new QSIC classification.

3. Method

3.1 Rational parameterization and bounded pullbacks

For the unit sphere, the rational stereographic map from the north pole is

```
phi^{-1}(u,v) = ( 2u/(1+u^2+v^2), 2v/(1+u^2+v^2), (1-u^2-v^2)/(1+u^2+v^2) ).
```

A general ellipsoid is handled by an affine map. For a quadric `Q`, define

```
F_{P,Q}(u,v) = Q(phi_P^{-1}(u,v)) * (1+u^2+v^2)^2,
```

a polynomial of degree at most four. Its leading homogeneous part is `Q(P) * (u^2+v^2)^2`, where `P` is the excluded point (the chart pole). We use "chart pole" and "excluded point" interchangeably. If `Q(P) != 0`, the real zero set is bounded. This is the boundedness reduction; the analytic core is formalized in Mathlib.

3.2 Deterministic multi-chart atlas

The atlas has six axial poles and twenty-four general poles with coordinates `(+/-1,+/-2,+/-2)/3` and permutations. A pole is safe if `Q_i(P) != 0` for every `i`. The kernel tries the atlas in a fixed order and uses the first safe chart that yields a certified 2D arrangement. If no chart is safe, the result is `NonBoundedChart`.

The atlas is not a completeness theorem. On a random benchmark a common safe pole exists in 2000/2000 two-to-three-plane cases; a synthetic worst case with 30 covering planes has zero safe poles. The residual all-chart rejection rate over 5000 fixed-seed plane cases is 14.98% at `H=1`, 0.64% at `H=3`, 0.020% at `H=10`, and 0.000% at `H=20`; this is empirical.

Exact 2D chart transforms. When a safe chart still produces an ill-conditioned pullback, the kernel may compose it with one of a fixed list of exact invertible 2D maps `M`: rational rotations `(3/5,4/5)` and `(5/13,12/13)`, shears `(1,1;0,1)` and `(1,0;1,1)`, and three determinant-`+/-2` maps. The transformed curve is `F o M`; the 2D topology is unchanged because `M` is an affine homeomorphism. The same `M` maps face points and vertices back to the base chart before 3D lifting, and each candidate must pass both the normal and strict certificates. The list is deterministic and fixed, not tuned per input; it is a tractability fallback, not part of the completeness argument. We deliberately do not order it by degree or coefficient size: an input-dependent order would make the accepted set depend on a heuristic, whereas the fixed order keeps two runs comparable.

3.3 Two-dimensional algebraic curve arrangement

The 2D core computes critical points from `F = F_v = 0`, pairwise intersections from resultants and subresultants, events and slabs, x-monotone arcs, face cycles, and face sign vectors at rational sample points. Real algebraic numbers are represented by a defining polynomial and an isolating interval (`AlgRoot`). Comparisons refine intervals until they separate; cost guards bound algebraic multiplication/addition. Every reported root carries an isolating interval, but we do not claim a separation bound that would guarantee that every event is found: the budget bounds work, not correctness, and a missed event remains a failure mode outside the certificate (Section 4.3, OP1).

Deterministic survey budget. To make the resource bound reproducible and independent of wall-clock time, the survey carries a `SurveyBudget`: maximum input and intermediate coefficient bits, maximum resultant matrix dimension, maximum root-isolation work units, maximum fiber refinements, and maximum `AlgRoot` comparison refinements. The budget is checked inside the Bareiss elimination, before square-free/GCD operations, in root isolation, and in the branch classifier. Exceeding it yields `BudgetExceeded` (fail closed), never a partial arrangement and never `CommonComponent`. The public `planar_arrangement_n` uses a wide default; the search uses a tight budget for three-or-more-surface primary candidates and wider budgets for transformed and basis candidates. The thresholds are engineering constants, not a complexity theorem.

Face sample points. Each face receives a rational sample point as follows. For an edge on a face cycle, take its slab and the branch index. The `u` coordinate is the rational midpoint of the slab, and the `v` coordinate is a rational point strictly between the branch and the next branch above/below (or between the branch and `+/-1` at the slab boundary). The point is therefore strictly inside the slab and strictly between two consecutive branches, hence inside the face; the kernel evaluates every curve at this point and rejects any zero. If no candidate point gives a nonzero sign vector, the arrangement fails closed.

3.4 Degeneracies and rejection certificates

Common components. A Gauss gate uses `content_v`, self-resultants, and pair resultants to detect shared components before the sweep. Detected cases are rejected as `CommonComponent`; the gate is sound. A separate fallback constructs a set-theoretic square-free coprime basis instead of rejecting the chart: it removes non-real `u`-only content, normalizes in `v`, and splits shared factors with a primitive PRS GCD. The basis has the same real zero set as the original curves. The 2D arrangement is built from the basis, but face labels are recomputed from the original quadrics, and the result is certified against the original pullbacks with both the normal and strict checks. This handles `self_resultant` charts without claiming support for true shared components. The division of labour is therefore explicit: the gate rejects on the original pullbacks (fast and sound) in the paths that do not build a basis, whereas the basis stage separates a shared or repeated factor `h` into its own curve (`f = h a`, `g = h b` gives `h, a, b`) and certifies against the original quadrics. The 14 `CommonComponent` counts in Section 5.2 come from the single-chart diagnostic, which does not invoke the basis stage.

Ordinary cusps. Let `x = u - u_c` and let `delta` be the offset from the repeated tangent direction; `T_3` is the third-order tangent derivative. For a point with `F=F_u=F_v=0`, discriminant `D=0`, `F_vv != 0`, and `T_3 != 0`, the two local branches satisfy

```
delta^2 = - x^3 T_3 / (3 F_vv^4).
```

Since `F_vv^4 > 0` and `3 > 0`, the sign is determined by `-x^3 T_3`. For `x>0`, `T_3<0` gives the right branch and `T_3>0` the left branch. The kernel merges the ordinary cusp into the fold-point path using this exact sign. The sign lemmas and the normal form are proved in Mathlib.

Exact membership. To decide `f(alpha,beta)=0` for independent algebraic numbers, the kernel uses univariate resultants, GCDs, and dynamic zero-divisor handling rather than bivariate GCD or algebraic-number arithmetic. Counterexamples from earlier prototypes are kept as regression tests.

Segre key. For pair classification, a `GL(2)`-invariant Segre key is computed with a non-singular gauge and univariate determinantal divisors. This key is used only for the handled cases; we do not claim a new QSIC classification.

3.5 3D assembly and the single-safe-chart lift

The 2D arcs are lifted to 3D edges; face cycles are assembled from the rotation system; each face carries a sign vector. For an ellipsoid plus three planes, the kernel closes and certifies 20/20 general-position cases with max `V/E/F = 12/18/8`.

For a scene with a common safe chart, the single-safe-chart path avoids 3D stitching: it lifts the certified 2D arrangement directly to `E`, adds the pole to the unbounded face, and uses the exact 2D labels. This handles empty quadrics (no real intersection with `E`) and self-intersecting space curves (case 17 has a figure-eight node) through the 2D pullback. The **3D-face-check mode** uses the same single chart but verifies face sample points and labels in 3D; it is a validation mode, not a multi-chart stitch. If neither mode accepts, the kernel returns a rejection. The certified API for planes covers up to six planes; seven or more are rejected as `OutOfScope` (a scope label, not a face-cycle failure). The gate is neither a geometric failure nor a budget cutoff: `BudgetExceeded` is reserved for exhaustion inside the deterministic `SurveyBudget` (Section 3.3), whereas the plane API stops at the largest corpus for which the same certificate has been validated (3/4

planes originally, 5/6 planes at 20/20 each). Returning `OutOfScope` keeps the two causes separate instead of reporting a resource exhaustion that did not occur.

Proposition 1 (single-chart lift). If `P` is a common safe pole (`Q_i(P) != 0` for all `i`) and the 2D arrangement of the pullbacks passes the exact 2D certificate, then its lift to `E`, with `P` added to the unbounded face, is a cell decomposition of `E` with exact signed labels. Proof sketch: the stereographic projection is a homeomorphism from `E \ {P}` to `R^2`; `P` lies on no curve; the unbounded face compactifies at `P`; labels are exact at rational face points and verified at 3D face points. Generalizing this to multi-chart stitching is future work. The 2D unbounded face is compactified at `P`; the 2D Euler relation `V-E+F=1+C` is preserved, and for a connected arrangement `C=1` this is the sphere value `2`. No extra vertex is added at `P`.

3.6 Determinism and cost

All combinatorial containers use `BTreeMap`/`BTreeSet` or explicitly sorted vectors. Random seeds are fixed. Floating point never decides a predicate. Artifacts are raw bytes with SHA-256 records, and CI checks byte stability. Costs are dominated by algebraic-number comparisons and resultants; the kernel uses cost guards and reports measured timings.

4. Soundness and Determinism

4.1 Exact layers and the floating-point boundary

The exact layer is `BigRational` (`Q`, rationals), `AlgRoot` (algebraic numbers with defining polynomial and isolating interval), and `PolyUV` (bivariate polynomials). Predicates are sign evaluation at rational points, exact algebraic comparison, nonzero tests for resultants and contents, polynomial GCDs, and membership tests. Floating point is used only for candidate generation and cost estimates.

4.2 Outcomes

The kernel has three outcomes:

1. `Accepted(Some(arrangement))`: a boundary complex with vertices, edges, faces, labels, and a rotation system.
2. `Rejected(None(reason))`: a sound rejection with a classified `NoneReason`.
3. `Undetermined`: a rejection whose reason is not yet classified.

4.3 Soundness of accepted outputs

Acceptance is not a claim of global completeness. The independent checker validates the face-cycle chain and half-edge pairing, Euler `V-E+F=1+C`, exact signs at rational face points, rotation coverage, and vertex metadata. Tampered labels, component counts, and rotation systems are rejected. A strict loop-boundary checker additionally evaluates non-incident curve signs at loop vertices with exact rational interval enclosures; it passes 20/20 on 3/4/5/6-plane and 2-quadric benchmarks, and 100/100 on the 3-quadric 100-case stress family. For extracted Rust data, Lean verifies closure and Euler on five scenes; negative controls are rejected. This is a certificate for accepted outputs, not a proof that the generator always accepts.

Proposition 2 (strict boundary certificate). Assume the arrangement is correct in the following structural sense: its edges are maximal arcs whose interiors meet `C` only at vertices, `F` is a connected component of the arrangement complement, and `C` is a union of arrangement edges. Suppose every boundary vertex of `F` not incident to `C` has the same nonzero sign `s`, and the face label for `C` is `s`. Then `F` lies in the open region `sign(C)=s`.

Proof sketch: under the structural assumption, an edge cannot cross `C` in its interior; it meets `C` only at incident vertices. Hence the sign of `C` is locally constant along the boundary away from incident vertices, and the boundary lies in the closure of the sign-`s` region. Since `F` is connected and disjoint from `C`, `F` is contained in that region.

The strict checker verifies the vertex-level consequent of this proposition on the constructed face cycles. It is a **post-hoc consistency check**, not a proof of the structural assumption. If every boundary vertex is incident to `C` (for example, a digon lens bounded by two curves), the non-incident set is empty and the exact face-point sign check decides the label. The Lean file `S5Strict.lean` formalizes the local checker; the geometric bridge and the edge-splitting assumption are paper-level and are not fully formalized (see OP1).

Checker detects	Relies on the structural premise
face-label / non-incident vertex sign mismatch	an edge interior crossing `C`
Euler / component-count mismatch	missed root isolation or event handling
rotation-system / open-edge mismatch	incorrect edge splitting
tampered face, label, or rotation system	global Jordan/connectedness

What the strict checker does not see. The checker is post-hoc: it assumes that the arrangement's edges are maximal arcs and that root isolation did not miss an event. If two curves form a tiny digon lens and both intersection vertices are missed, the resulting face cycle can still satisfy Euler and the vertex-level sign checks; the non-incident set may even be empty. This is why the structural premise (OP1) is stated explicitly and why the strict check is not a completeness proof for the arrangement generator.

Rational chart transforms. Let `M` be an invertible `2 x 2` rational matrix and `F' = F o M`. Since `M` is an affine homeomorphism of `R^2`, it induces a bijection on zero sets, vertices, arcs, faces, and face adjacency. The kernel enumerates the same combinatorial arrangement after applying `M^{-1}` to the stored coordinates; an accepted candidate must still pass the certificate against the transformed pullbacks.

Square-free coprime basis. Replacing a curve family by a square-free coprime basis preserves the union of real zero sets: removing a non-real `u`-only content factor removes no real point, and splitting a shared factor `h` from `f = h a`, `g = h b` replaces `{f=0} union {g=0}` by `{h=0} union {a=0} union {b=0}`, which is the same set. Labels are recomputed from the original quadrics, and the final arrangement is certified against the original pullbacks, so the basis is a construction aid rather than a change of semantics.

4.4 Determinism

The kernel uses deterministic containers, fixed seeds, canonical sorting, and no f64 decisions. Lean proves canonical order uniqueness, merge-sort determinism, label multiset invariance, and injective base-4 label encoding. Rust checks two-run equality and byte-stable artifacts; CI checks the same outputs across platforms.

4.5 What is proved, checked, measured, and assumed

- **Proved**: `checkMatch` iff, `openEdges` closure, label multiset/injective encoding, `NoneReason` totality, cusp branch sign, interval comparator order axioms, Euler parity/duality/tree, and the analytic boundedness core.
- **Checked**: five extracted Rust scenes for closure and Euler; negative controls.
- **Measured**: completeness, adversarial cases, external oracle, ablation, scale.
- **Assumed**: the full arrangement construction is not formally verified; the general planar genus-zero theorem is not formalized; the polynomial leading-term reduction and the general Taylor `O(4)` truncation are paper-level arguments.

The Lean development has no `sorryAx` in its main theorems; the two layers (Mathlib-free and Mathlib) are checked separately.

5. Completeness Evaluation

5.1 Two layers

The single-chart diagnostic fixes one chart and measures rejection; it isolates soundness and degeneracy handling. The multi-chart product path measures residual all-chart rejection and practical completeness. Mixing them would hide the atlas effect.

5.2 Single-chart diagnostic

The single-chart diagnostic rejects 378 inputs: 364 non-bounded charts and 14 confluent components, with zero unclassified. On product geometry, the circle family closes 8000/8000; the ellipsoid-plane family records `ok=1617`, `none=14`, all 14 explainable by a degenerate class, with zero general-position rejections.

The population is 10,000 generated cases: 8,000 random circle arrangements and 2,000 ellipsoid-plane scenes. After filtering (undefined ellipsoid, fewer than two curves, non-quadratic-in-`v` pullback, duplicate planes), 9,631 are attempted (8,000 circle and 1,631 ellipsoid-plane); the 378 rejections refer to this attempted set. The 369 filtered ellipsoid-plane cases are not counted as successes or failures.

5.3 Atlas residual and ablation

The atlas has six axial and twenty-four general poles. On 5000 fixed-seed cases per coefficient height `H`, the residual all-chart rejection rate is below. Each case uses two or three random planes on a random rational ellipsoid: this is a plane-family benchmark, not a general-quadric benchmark. The mixed quartic family is evaluated separately.

H	cases	None	rate
1	5000	749	14.98%
3	5000	32	0.64%
10	5000	1	0.020%
20	5000	0	0.000%

The trend is consistent with an empirical `O(1/H)` behavior, not a theorem. On 2000 two-to-three-plane cases a common safe pole exists in all 2000 (average 25.64 of 30 poles safe); a synthetic worst case with 30 covering planes has zero safe poles.

The ablation isolates chart switching. The plane `x=0` is non-bounded in the base chart but the atlas resolves it (`V/E/F=2/2/2`). Two vertical planes are non-bounded in the base chart and are resolved (`V/E/F=6/8/4`). A plane tangent at the excluded pole is rejected by the atlas as well (fail-closed).

5.4 Quartic pullbacks and single-safe-chart lift

For the 20 mixed quadric pullbacks at `H=2`, the standard chart closes 14. The six failures are all non-bounded; the undetermined count is zero. Appendix A gives the per-case details: degree, degree in `v`, boundedness, singularity, and separate `detected` and `reject_reason` fields. A row with `result=ok` may carry a `detected` tag (e.g., `singular-point`) that is distinct from `reject_reason`; a detected singularity that is resolved is not a rejection.

The 3D-face-check mode uses the same 6+24 candidate pool as the single-chart path. An earlier version closed 17/20: cases 2 and 17 were lost and case 13 was unresolved because an empty quadric aborted the whole search and a common-component chart aborted its search. After making these gates per-chart (skip an empty quadric; continue past a common-component chart) and certifying each accepted candidate, the 3D-face-check mode closes **20/20 certified**. Case 2 has one empty quadric, case 13 switches away from a pole on the cylinder, and case 17 has a figure-eight node; all three are handled by the same candidate pool. This mode still reconstructs **one** chart per output; it does not stitch different charts.

Table 4b. Quartic closing rates by method (2-quadric 20-case benchmark).

method	closes	note
standard chart	14/20	6 non-bounded
6+24 chart search + 3D face-point check	**20/20**	certified; one chart per output
single safe chart 2D lift	**20/20**	certified; strict 20/20
last-resort 3D check	**0/20**	not invoked on this benchmark

All rows use the same 6+24 candidate pool; the difference is the reconstruction and acceptance check. The single-safe-chart 2D lift is preferred because it uses the exact 2D labels and compactifies the unbounded face at the pole. Genuine multi-chart stitching is not implemented (OP2).

A **20-case three-quadric pilot** (same seed family, `H=2`) is the first test with triple-point potential: 12/20 close via the primary single-chart path and 8/20 via transform/basis fallbacks; all 20 are certified and strict, and the last-resort 3D check is 0/20. The earlier 18/2 split disappeared after removing a duplicate chart-preparation pass: the old code zipped charts sorted by the general key with keys sorted by the plane key, so an all-plane case could pair a chart with the wrong priority key. The corrected path uses `chart` and `key` from the same prepared entry.

A **100-case three-quadric stress test** (8 parallel workers, 300 s per case) is the larger check. With the deterministic survey budget, exact 2D chart transforms, and the basis fallback, it closes **100/100**, certified **100/100**, strict **100/100**, with no timeout. The former failures 31 (`cone+cone+cylinder`) and 66 (`cylinder+cylinder+plane`) and the former timeouts 12 and 57 all pass. In this family 61 cases use the primary single-chart path, 37 require only a rational linear transform, and 2 require the square-free coprime basis; the basis+linear and last-resort 3D stages are not needed. This is still an empirical result on a fixed-seed, `H=2` family, not an unconditional three-quadric theorem.

Table 4c. Three-quadric 100-case stress (8 parallel workers, 300 s/case).

method	closes	certified	strict	timeout
hybrid (single + transform/basis fallback)	**100/100**	**100/100**	**100/100**	**0**
primary single-chart only	61/100	—	—	—
transform/basis fallback	39/100	—	—	—

Latency in this 8-worker run: median 9.2 s, maximum 104.0 s; 13 cases exceed 30 s, three exceed 60 s, and none exceeds 120 s. The tail is dominated by cases that try several transformed charts before a strict-valid candidate. Moving the det-2 shear earlier reduced case 57 from 126 s to 64 s in the same harness. An additional early exit — if `Res_v(F,G)` has no real root, the pair cannot intersect, so `Res_u` and the `v` root isolation are skipped — reduced case 38 from 88 s to 31 s single-job. A sequential 20-case pilot gives median 6.3 s and maximum 31.7 s.

Table 4d. Fallback attribution for the 100-case stress.

stage that produced the accepted output	cases	notes
primary single-chart	61	includes cases with intermediate budget pruning
rational 2D linear transform	37	rotations and shears; 12/38/57/79/99 are examples
square-free coprime basis	2	31 and 66
basis + linear transform	0	not needed
last-resort 3D check	0	not needed

The diagnostic `SurveyBudget` counter records 9,651 rejected intermediate operations across 82 cases, but **no case failed solely because of `BudgetExceeded`**; the counter prunes candidate resultants/roots and the search continues. Cases 31 and 66 need the basis (0 budget hits); the former timeouts 12 and 57 need the linear transform. This addresses the natural question of whether the 100/100 result is just a budget cutoff: it is not.

The attribution is not a monotone ablation, because each stage is a complete search rather than a filter; it does show that neither the linear stage (37 cases) nor the basis stage (2 cases) is redundant on this family. Basis+linear and the last-resort 3D check contribute zero accepted outputs, so the 100/100 result is obtained by the first three stages alone.

Table 4e. Budget/transform sensitivity pilot. `P64`…`P1024` are the primary single-chart budget in resultant bits; `L` / `Lrev` are the linear stage with forward/reverse transform order; `B` is the basis stage. A checkmark means the stage produced a certified+strict output within the run; seconds are wall-clock in that run.

case	P64	P128	P256	P1024	L	Lrev	B	accepted stage
0-3 (pilot)	ok	ok	ok	ok	ok	ok	ok	primary
12	fail	fail	ok (76 s)	ok (75 s)	ok (27 s)	ok (69 s)	ok (79 s)	linear
31	fail	fail	fail	fail	fail	fail	ok (49 s)	basis
66	fail	fail	fail	fail	fail	fail	ok (0.35 s)	basis

Two conclusions are already visible. First, the budget is not a binary success switch: case 12 needs at least 256 bits in the primary path but is solved faster by the linear stage at 64 bits. Case 31/66 are budget-insensitive and need the basis. Second, transform order affects latency (27 s vs 69 s on case 12) but not the accepted set in this pilot; case 57 needs the det-2 shear and is too slow for the full 7-way ablation; we report its accepted path from Table 4d and the 126 s -> 64 s improvement from moving that shear earlier.

5.5 External oracle and certificate checks

We cross-check intersection counts on six fixed cases against SymPy: general (2), external tangency (1), internal tangency (1), concentric (0), triple point (4), and quartic plus conic (4). The Rust counts match. The certificate checker accepts representative arrangements and rejects tampered labels, component counts, and rotation systems.

5.6 Scale and negative results

The scale probe over circle arrangements `N=2..7` closes all cases with the certificate passing; wall times range from 1.9 ms to 167 ms. These numbers demonstrate practical behavior on the tested family, not a general complexity bound.

The 20 two-quadric cases take 0.03–7.1 s end-to-end including certificate and strict checks (median 2.3 s, total 49.0 s on the reference machine, one job). Removing the duplicate chart-preparation pass cut the earlier end-to-end median from 3.9 s to 2.2 s. The 100-case three-quadric sweep no longer times out; its latency tail is reported in Table 4c.

A stage profile after that fix attributes **46% to chart preparation, 52% to 2D arrangement, and under 1% to the 2D certificate, 3D lift, full certificate, and strict check combined**. The end-to-end gap is therefore chart search/pullback and 2D construction, not strict post-hoc checking. This is why the quartic benchmark stays at 20 cases and why the three-quadric run is a pilot rather than a large corpus.

The four-plane case is resolved by the same single-chart lift: 4-plane **20/20** certified (example C1 `V/E/F=20/32/14`). The earlier shared face-cycle check produced mixed boundary signs at rationally approximated 3D vertices; this was a checker false positive, not a topology error. A strict loop-boundary check (exact interval evaluation at 2D vertices) passes 20/20 on 3/4/5/6-plane and quartic benchmarks; a general proof of the strict check remains future work. The certified claim uses exact 2D face points, exact 3D face-point signs, and the benchmark-verified strict check. Seven or more planes are out of scope. Four general quadrics remain out of scope. Mesh fidelity is best-effort.

6. 3D Arrangement and Application

6.1 Ellipsoid plus three to six planes

For an ellipsoid plus three planes in 20 general-position cases, the certified kernel closes 20/20 with max `V/E/F=12/18/8`. Four planes also close 20/20 (example C1: `V/E/F=20/32/14`). The same single-chart lift covers five and six planes: both families close **20/20** certified and strict, with max `V/E/F=30/50/22` (5 planes) and `42/72/32` (6 planes). Per-case latency over these 20-case families is 36.8 ms (3 planes), 75.0 ms (4), 126.5 ms (5), and 185.3 ms (6) median; the maximum is 245 ms and peak process working set is about 6.9 MB. Every face point lies on the ellipsoid, every face label matches the exact plane signs, and two runs agree. Plane-ellipsoid intersections are circles (conics) in the pullback, so this family is a **baseline** for the pipeline and face labels, not evidence for general quartic 3D; the quartic single-chart lift is the relevant test. The certified plane API explicitly gates `planes.len() > 6` and returns `OutOfScope`; this is a validation-scope label, not a face-cycle failure and not a budget cutoff (the distinction is explained in Section 3.5). The single-chart lift itself is n-agnostic; seven or more planes are outside the validated corpus.

6.2 Band scene

The band scene is the unit sphere cut by two parallel planes `z=+1/2` and `z=-1/2`; the central face is an annulus with two loops. It closes with 64 triangles, half-edge pairing `(0,0)`, exact counts `(64,0,0)`, and zero self-intersections. It is a baseline for the exact manifold path; the mesh is not claimed to be optimal or seam-free.

6.3 Auxiliary 2D slicer

The kernel can drive an exact 2D slicer: boundary arcs are split at exact intersection points and can be exported to SVG or DXF. This is auxiliary and not used to support the main evaluation.

6.4 Certificate checker

The checker is the bridge between "the kernel returned something" and "the returned object is independently checkable." It is deliberately small and separate from the generator. Its negative tests are as important as its positive tests. The checker does not prove that the generator is complete; it proves that accepted outputs satisfy the stated invariants. A two-phase API prototype separates Phase 1 (fast topology and non-strict certificate) from Phase 2 (strict loop-boundary check); Phase 1 is for preview/topology consumers, and Phase 2 is the audit gate.

7. Limitations

- **No unconditional completeness.** The kernel can return `Rejected(None(reason))` or `Undetermined`; the claim is soundness plus empirical practical completeness.
- **Seven or more planes are out of scope.** 3/4/5/6-plane families are certified 20/20 under the scoped certificate and strict check. The earlier shared face-cycle check was a false positive due to rationally approximated vertices; the certified claim uses exact 2D face points and exact 3D face-point signs.
- **Four general quadrics are out of scope.**
- **Quartic pullbacks:** the 2-quadric benchmark is standard 14/20; the 3D-face-check mode is 20/20 certified; the single-safe-chart 2D lift is 20/20 certified, with last-resort 3D check 0/20. The 3-quadric pilot and the 100-case stress test are 100/100 certified and strict, with no timeout under the 300 s/case harness; 39/100 use a transform or basis fallback. This is still an empirical fixed-seed result, not a general three-quadric claim. The lift requires a common safe chart or a transform that makes the chart tractable; without either, true multi-chart stitching remains future work (OP2).
- `O(1/H)` **is empirical**, measured on the 2–3 plane family with a fixed seed; it is not a theorem.

- **No head-to-head benchmark.** We do not compare wall-clock time or memory against CGAL or other exact arrangement implementations; the evaluation is internal and fixed-seed.
- **Mesh fidelity is best-effort**; arc-seam artifacts remain.
- **Lean coverage is partial**; the full arrangement construction is not verified.

7.1 Open problems

ID	Problem	Current status	Remaining work
OP1	Strict loop-boundary certificate for 3D face cycles	local checker Lean-formalized; passes 20/20 on 3/4/5/6-plane and quartic benchmarks	geometric bridge/edge-splitting proof and full formalization
OP2	Multi-chart stitching without a common safe chart	single chart preferred; 3D-face-check limited; true multi-chart stitching unimplemented	arbitrary-scene stitching and completeness
OP3	General four quadrics / seven-plus surfaces	out of scope	fourth-degree space curves, triple/quadruple points
OP4	Unconditional termination and cost bounds	resource limits / `Undetermined` possible	termination and cost theorem for the class
OP5	Lean coverage	combinatorial core + extracted data + Mathlib lemmas	extend coverage; full formalization is not planned
OP6	Performance and scaling	20/5000-case benchmark, small scenes	CAD-scale scenes, comparison with certified MT

We avoid phrases such as "solves all inputs," "complete for arbitrary degeneracies," "guarantees watertight meshes," and "formally verified arrangement construction."

8. Related Work

Prior work has established exact 3D quadric arrangements [Mourrain-Tecourt-Teillaud 2005], complete planar maps of quadric intersection curves [Berberich et al. 2005], exact cell computation including degenerate cases [Schomer-Wolpert 2006], exact low-degree curve arrangements and conic booleans [Exacus 2005], general exact arrangement frameworks [CGAL], exact boundary evaluation [ESOLID], isotopy-guaranteed implicit meshing [Plantinga-Vegter 2004/2007], and a complete QSIC classification via signature sequences [Tu et al. 2009]. Near-optimal parameterization of quadric intersections is addressed by Dupont et al., exact mesh CSG by Levy (2025), and CGAL's `Arrangement_on_surface_2` (including surface traits); the verified 3D mesh-intersection work is available as an open repository (schildep, 2026). We do not claim a new arrangement algorithm or a new classification.

Exact arrangements of quartic curves and implicitly defined real algebraic plane curves are treated by Caravantes and Gonzalez-Vega [2007, 2008], and complete arrangements of cubic curves with all degeneracies by Eigenwillig et al. [2006]. Our C4 benchmark is a scoped evaluation of a quadric-pullback pipeline, not a new quartic arrangement algorithm. The specific contribution of C5 is a specific exact membership test, not the general avoidance of irrational numbers, which is already present in [Caravantes–Gonzalez-Vega 2008].

Arrangements on parametric surfaces and exact spherical arrangements are established by Berberich et al. [2007, 2010], Cazals and Loriot [2009], de Castro et al. [2009], and Fogel [2009]. The classical stereographic parameterization of quadrics and its rational reparameterizations are analyzed by Wang, Joe, and Goldman [1997]. Our rational atlas uses these classical charts; the delta is the deterministic 6+24 pole atlas, the interval-Möbius vertex identity, and the empirical `O(1/H)` residual.

Formal-methods CSG is explored by Tzoumas et al. [2015], who apply interval analysis and proof assistants to CSG models. Certifying algorithms and untrusted-certificate architectures are established by McConnell et al. [2011] and Li, Passmore, and Paulson [2019]. Our contribution is the specific checker for Euler/face signs/rotation and the quadric-arrangement application, not the general idea of certification.

Separation bounds for real algebraic expressions and root comparison are classical [Burnikel et al. 2009; Emiris–Mourrain–Tsigaridas 2010]. Formal planar subdivisions, Euler formula, and discrete Jordan curve theorem are formalized with combinatorial hypermaps by Dufourd [2008, 2009]. Our Lean layer is partial and targets the half-edge/closure/Euler bookkeeping, not a full planar-map formalization.

A companion technical report (S4) covers the sphere-plus-plane specialization and its Lean combinatorial closure. S5 is not a superset of that report: it adds general quadric pullbacks, the 6+24 atlas, ordinary-cusp handling, the certified 3-6-plane gate, and the single-safe-chart lift. The shared Lean core is the half-edge/closure/Euler bookkeeping; S5-specific Lean is `S5Strict.lean`, the Mathlib cusp/boundedness lemmas, and the quartic extracted-data checks.

Our difference is the combination of guarantees: a rational multi-chart atlas for unbounded pullbacks, a classified `NoneReason` for every rejection, an independent certificate checker, deterministic and reproducible artifacts, and a partially formalized combinatorial core. ESOLID explicitly does not support degenerate Boolean configurations; we make degeneracy handling explicit, but our scope is narrower. The Lean-verified mesh-intersection work provides stronger end-to-end verification for triangle meshes; our target is quadric arrangements and our Lean coverage is partial.

The positioning is deliberate: compare us on the guarantees we actually provide, not on a claim of general superiority.

9. Reproducibility

- **Artifacts.** `data/artifacts/` contains raw stdout and SHA-256 records. Quartic 2-quadric benchmark: standard 14/20, 3D-face-check mode 20/20 certified, single-chart lift 20/20 certified/strict, fallback 0/20. Three-quadric pilot: hybrid 20/20 certified/strict; the 100-case stress test is 100/100 certified/strict with no timeout. Plane families: 3/4/5/6-plane 20/20 certified/strict, median 36.8/75.0/126.5/185.3 ms. The P1/P2 record the 100-case stress test is 100/100 certified/strict with no timeout. Plane families: 3/4/5/6-plane 20/20 certified/strict, median 36.8/75.0/126.5/185.3 ms. The P1/P2 results are summarised in `data/artifacts/s5_p1p2_results.csv`; the sweeps are `s5_quadrics_hybrid_3q_100_p1p2_2026-10-24.txt`, `s5_quadrics_hybrid_2q_20_seq_p1p2_2026-10-24.txt`, and `s5_quadrics_hybrid_3q_100_strategy_2026-10-24.txt`; the stage profile is `s5_quadrics_profile_2026-10-24.txt`. `SHA256SUMS` covers every artifact in this record.
- **Rust tests.** `s5_certificate`, `s5_adversarial`, `s5_sympy_oracle`, `s5_ablation`, `s5_horizontal`, `s5_arrangement`, `s5_curve2d`, `s5_quadrics_hybrid`, `s5_two_phase`, `fuzz_s5`. Correctness sweeps use the parallel harness; latency is measured with one job.
- **Lean.** The Mathlib-free core and the Rust-to-Lean extraction are checked for `sorryAx`, and the Mathlib files are checked separately. The Lean sources are not included in this artifact.
- **CI.** Ubuntu, macOS x86, macOS arm64, and Windows; byte-stable artifacts and tests.
- **Environment.** AMD Ryzen 7 5700X (8 cores / 16 threads), 64 GB RAM, Windows 11, rustc 1.98.0, release profile. Timings are indicative.
- **Randomness.** Fixed-seed xorshift64*; mixed quartic coefficients uniform over integer ranges determined by `H`; the `H` benchmark uses two or three planes with integer normals and offsets on a random rational ellipsoid.

9.1 Benchmark protocol

- Mixed quartic family: fixed seed, `H=2`, two quadrics per case, 20 cases.
- Atlas family: 5000 fixed-seed cases per `H`, two or three planes per case.
- 3/4/5/6-plane families: 20 deterministic cases each.
- Failure corpus: seven or more planes, general four quadrics, and strict-check cases.

- Inputs are deterministic and fixed-seed; the reported counts are recorded in the artifact data so the same benchmark can be reconstructed.

This artifact includes the artifact data and the SHA-256 list. The paper does not rely on a proprietary dataset. The research kernel and proofs are independent of the Axiom product QA core; the product boundary is described in Section 9.2.

9.2 Code and data availability

Code. The S5 research kernel, tests, examples, and Lean/Mathlib proofs are not included in this paper/data artifact. No code license is granted by this artifact.

Product boundary. The Axiom product QA core, MCP server, CLI, and CA signing service are separate from this paper and are not included in this artifact. Their licensing is separate and is not determined by this artifact. No proprietary A' source or CA signing key is included. None of the S5 results depend on the product.

Data. All benchmark families are synthetic and fixed-seed: the 2-3-plane atlas family (5000 cases per `H`), the 2-quadric 20-case family, the 3-quadric 20/100-case families, and the 3/4/5/6-plane 20-case families. Inputs are deterministic. No proprietary, personal, or third-party restricted data are used. SymPy is an optional independent oracle; CGAL is not used for S5 claims.

Certificate and verification. The S5 certificate is the runtime checker `certify_planar_arrangement_n` (and its strict variant); its tests include four accepted arrangements and three tampered arrangements that must be rejected. The checker construction is described in this paper; its implementation is not included in this artifact. The Axiom-signed certificate schema is open. Axiom-signed certificates are not used as S5 evidence.

License. Paper text: CC BY 4.0. Artifact data: CC BY 4.0. Third-party tools keep their own licenses. No research code, Lean source, product source, MCP server, CLI, or CA signing key is included in this artifact.

9.3 Reproducibility scope

All reported algorithmic results and benchmark counts are recorded in this data artifact: the atlas rejection counts, the 2-quadric 20/20 and 3-quadric 100/100 certified/strict results, the 3/4/5/6-plane 20/20 results, the SymPy cross-checks, and the Lean checks. Timing, parallel speed-up, and peak-RSS values depend on the host and worker count and are not bit-reproducible. Axiom-signed certificates and the Axiom product QA core are outside the reproducibility scope of S5; no S5 claim depends on them. Test vectors are included under `data/artifacts/test-vectors/`.

Appendix A. Per-case quartic details

Table 4. Per-case quartic details. `curve1` and `curve2` report degree / degree in `v` / boundedness / singularity. `detected` is a tag on an accepted case; `reject_reason` is a `NoneReason` on a rejected case. Legend: B/U = bounded/unbounded, S/NS = singular/non-singular, NB = NonBoundedChart, sing = singular-point.

case	kinds	result	detected	reject_reason	curve1	curve2
0	cylinder+plane	none	-	NB	3/3/U/NS	2/2/B/NS
1	cylinder+cylinder	none	-	NB	4/4/B/NS	3/3/U/NS
2	cone+general	ok	-	-	4/4/B/NS	4/4/B/NS
3	cone+plane	ok	-	-	2/2/B/NS	4/4/B/NS
4	cylinder+cylinder	none	-	NB	4/4/B/NS	3/2/U/NS
5	general+plane	ok	-	-	4/4/B/NS	2/2/B/NS
6	cone+general	ok	-	-	4/4/B/NS	4/4/B/NS
7	cone+plane	ok	sing	-	2/2/B/NS	4/4/B/S
8	cone+cylinder	ok	sing	-	2/2/B/S	4/4/B/NS
9	cone+cone	ok	sing	-	4/4/B/NS	4/4/B/S
10	general+general	ok	-	-	4/4/B/NS	4/4/B/NS
11	plane+plane	none	-	NB	1/1/U/NS	2/2/B/NS
12	cone+plane	ok	-	-	2/2/B/NS	4/4/B/NS
13	cylinder+general	none	-	NB	4/4/B/NS	3/3/U/NS
14	cone+cone	ok	-	-	4/4/B/NS	4/4/B/NS
15	cylinder+general	ok	-	-	4/4/B/NS	4/4/B/NS
16	general+plane	ok	-	-	2/2/B/NS	4/4/B/NS
17	cylinder+general	ok	sing	-	4/4/B/NS	4/4/B/S
18	cylinder+plane	none	-	NB	4/4/B/S	1/1/U/NS
19	cone+general	ok	-	-	4/4/B/NS	4/4/B/NS

This table is the 2-quadric benchmark: the 3D-face-check mode and the single-safe-chart 2D lift both close all 20, and the last-resort 3D check is not invoked (0/20). The separate 3-quadric pilot is hybrid 20/20 certified/strict. The 100-case stress test is not shown per case; it closes 100/100 with 100/100 certified and strict and no timeout. Cases 31, 66, 12, and 57, which failed or timed out before the deterministic-budget and transform/basis work, all pass.

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